ECEN 5692 HW1 Matt Hartnett

Problem 0.1.A.1:

B - No

Since the spin of electron A is random with an equal liklihood of up or down, the receiver observing electron B cannot gain any information, tho there is no communication.

Problem 1.2.A.3:

$f_c = 500\ MHz$

$\lambda = c/f$

$\approx 0.60m$

$\frac{1}{2}\lambda = \boxed{0.3m \text{ antenna size}}$

Problem 1.2.A.4:

Baseband $x(t) = -5\sin(2\pi 5t) + j3\cos(2\pi 10t)$

$f_c = 1.9\ GHz$

1. $I/Q\uparrow(x(t)) = R_e[x(t)\sqrt{2}\, e^{j2\pi f_c t}]$

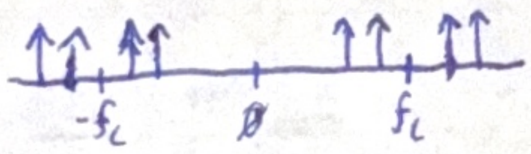
$= \sqrt{2}\, R_e[(-5\sin(2\pi 5t) + j3\cos(2\pi 10t)e^{j2\pi f_c t}]$

$= \sqrt{2}\, R_e[(-5\sin(2\pi 5t) + j3\cos(2\pi 10t)(\cos(2\pi f_c t) + j\sin(2\pi f_c t)]$

Only real terms matter, so after multiplication...

$= \sqrt{2}\,[-5\sin(2\pi 5t)\cos(2\pi f_c t) - 3\cos(2\pi 10t)\sin(2\pi f_c t)]$

$= \sqrt{2}\,[-\frac{5}{2}(\sin(2\pi t(f_c+5) + \sin(2\pi t(f_c-5))) - \frac{3}{2}(\sin(2\pi t(f_c+10)) + \sin(2\pi t(f_c-10))]$

2. 

$f_i = [-10, -5, 5, 10]$ since there are 4 spikes around each carrier frequency.

$\tilde{x}_c(f) = I/Q\uparrow(\tilde{x}(f))$

$= \dfrac{\tilde{x}(f-f_c) + \tilde{x}^*(-f-f_c)}{\sqrt{2}}$

$\tilde{x}(f) = \mathcal{F}\{-5\sin(2\pi 5t) + j3\cos(2\pi 10t)\}$

$= -5\mathcal{F}\{\frac{1}{2j}(e^{j2\pi 5t} - e^{-j2\pi 5t})\} + j3\mathcal{F}\{\frac{1}{2}(e^{j2\pi 10t} + e^{-j2\pi 10t})\}$

$= -5/2j\,(\delta(f-5) - \delta(f+5)) + j3/2\,(\delta(f-10) + \delta(f+10))$

~~$f_i = [-10, -5, 5, 10]$~~

~~$a_i = [3/2, j, -5/2j$~~

$$\boxed{\begin{aligned} f_i &= [-10, -5, 5, 10] + 1.9\times10^9 \\ a_i &= [3j/2, -5/2j, 5/2j, 3j/2] \end{aligned}}$$

~~$3j/2, 5/2j, 3j/2$~~

ECEN 5692 HW1 Matt Hartnett

Problem 1.2.A.5:

Baseband $x(t) = 4\cos(2\pi 7t)$

$f_c = 1.9$ GHz

1. $I/Q\uparrow(x(t)) = Re[x(t)\sqrt{2}\, e^{j2\pi f_c t}]$

$$X(t) = 4\left(\frac{e^{j2\pi 7t} + e^{-j2\pi 7t}}{2}\right)$$

$$X_c(t) = Re[\sqrt{2}\cdot 2(e^{j2\pi 7t} + e^{-j2\pi 7t})e^{j2\pi f_c t}]$$

$$= 2\sqrt{2}\, Re[e^{j2\pi(f_c+7)t} + e^{-j2\pi(f-7)t}]$$

$$\boxed{X_c(t) = 2\sqrt{2}(\cos(2\pi(1.9\times10^9+7)t) + \cos(2\pi(1.9\times10^9-7)t))}$$

2. $f_i = [-7, 7] + 1.9\times10^9$

~~We can start w/~~

$$\tilde{X}(f) = 4\cdot\frac{1}{2}(\delta(f-7) + \delta(f+7))$$

$$\tilde{X}^*(-f-f_c) = \tilde{X}(f+f_c)$$

$$\tilde{X}_c(f) = \frac{\tilde{X}(f-f_c) + \tilde{X}^*(-f-f_c)}{\sqrt{2}}$$

$$= 2/\sqrt{2}\,(\{\delta(f-(f_c+7)) + \delta(f-(f_c-7))\} + (\delta(f+(f_c-7)) + \delta(f+(f_c+7)))$$

$$\boxed{\begin{aligned} f_i &= [1.9\times10^9-7,\ 1.9\times10^9+7] \\ a_i &= [2/\sqrt{2},\ 2/\sqrt{2}] \end{aligned}}$$